Online Supplement: Carbon-Aware Scheduling for Phase-Disaggregated LLM Inference in Cloud and Serverless Platforms

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Revision v2 – Artifact Supplement

Code and Reproducibility Repository: https://github.com/UmarIqbal000/carbon-aware-llm-scheduling (SimPy 4.1.2, paired seeds 100-109, runner experiments/reproduce_all.py).

S1. PRISMA-2020 V2 Corpus Reconciliation Summary

Table 1 summarizes the updated V2 screening accounting logged in research/search_log.md, expanding the V1 corpus ($n = 47$) with 23 newly screened V2 candidate records (9 explicitly audited anchors plus 14 additional 2024–Nov 2026 search hits), removing 2 duplicates, excluding 3 items at title/abstract and 4 items at full-text eligibility, and retaining $n = 61$ unique works.

Table 1: Updated V2 Corpus Characteristics ($n = 61$ Included Works)

Dimension	Verified Count and Composition
Screening Flow	98 identified → 3 duplicates removed → 95 screened → 14 title/abstract excluded → 81 assessed → 20 full-text excluded → **61 included**.
Source Channels	Channel 1 (surveys): 9; Channel 2 (bounded API queries): 14; Channel 3 (targeted anchors, benchmarks, standards): 38 ($9 + 14 + 38 = 61$).
Thematic Pillars	Pillar (a) surveys/agendas: 11; Pillar (b) carbon cloud/serverless/geo: 15; Pillar (c) P/D & LLM serving: 21; Pillar (d) embodied/traces/standards: 12; Pillar (e) foundations: 2.
Publication Recency	56/61 from 2022–2026 (**91.8%**); 34/61 from 2025–2026 (**55.7%**); exact counts: 2011 (2), 2020 (1), 2021 (2), 2022 (1), 2023 (6), 2024 (15), 2025 (16), 2026 (18).
Audit Depth	56/61 full-text audited (**91.8%**); 5/61 abstract-only (**8.2%**).
Preprint Status	16 preprints total: 15 arXiv-only preprints plus 1 accepted ATC'26 conference preprint (wang2026festina, marked % TODO-VERIFY pending Crossref DOI indexing).

S2. Complete 24-Check Literature Anchor Simulator Validation

All 24 quantitative checks generated by experiments/validate_simulator.py (saved in validation_benchmarks.csv) appear in Table 2. Across uncontended single-request Splitwise Table IV points, simulated TTFT includes parameterized KV stall overhead (+5.0 ms on H100, +8.0 ms on A100), explaining the +8.81% to +14.24% positive offset above raw Splitwise compute latencies. Under power capping, a 0.55× TDP decode cap leaves 16-request decode step latency flat at

29.25 ms while dropping node power from 1563.5 W to 1069.8 W (−31.6%), before an abrupt +6.0% step penalty appears at 0.40× TDP (31.01 ms), matching Splitwise Figs. 8–9.

Table 2: Complete 24-Check Simulator Anchor Validation (`validation_benchmarks.csv`)

Suite	Condition	Sim	Ref	Err (%)	Notes
Splitwise Table IV (Conv)	H100 ($L_{\text{in}} = 1113$), TTFT (ms)	95.96	84.0	+14.24	raw=84.0 + kv_stall=5.0 ms
Splitwise Table IV (Conv)	H100 ($L_{\text{in}} = 1113$), TPOT (ms)	28.00	28.0	0.00	0.55 TDP preserves TBT
Splitwise Table IV (Conv)	A100 ($L_{\text{in}} = 1083$), TTFT (ms)	169.73	155.0	+9.50	raw=155.0 + kv_stall=8.0 ms
Splitwise Table IV (Conv)	A100 ($L_{\text{in}} = 1083$), TPOT (ms)	42.00	40.0	+5.00	0.55 TDP preserves TBT
Splitwise Table IV (Code)	H100 ($L_{\text{in}} = 1290$), TTFT (ms)	108.04	95.0	+13.73	raw=95.0 + kv_stall=5.0 ms
Splitwise Table IV (Code)	H100 ($L_{\text{in}} = 1290$), TPOT (ms)	28.00	31.0	-9.68	0.55 TDP preserves TBT
Splitwise Table IV (Code)	A100 ($L_{\text{in}} = 1333$), TTFT (ms)	201.29	185.0	+8.81	raw=185.0 + kv_stall=8.0 ms
Splitwise Table IV (Code)	A100 ($L_{\text{in}} = 1333$), TPOT (ms)	42.00	52.0	-19.23	0.55 TDP preserves TBT
Splitwise Figs 8–9	H100 batch-16, cap=1.00 TDP	29.25	28.0	+4.47	Power = 1563.5 W
Splitwise Figs 8–9	H100 batch-16, cap=0.70 TDP	29.25	28.0	+4.47	Power = 1234.4 W
Splitwise Figs 8–9	H100 batch-16, cap=0.55 TDP	29.25	28.0	+4.47	Power = 1069.8 W
Splitwise Figs 8–9	H100 batch-16, cap=0.50 TDP	29.25	28.0	+4.47	Power = 1015.0 W
Splitwise Figs 8–9	H100 batch-16, cap=0.40 TDP	31.01	–	–	Power = 905.3 W (+6.0% step)
ServerlessLLM Fig 6a	Llama-2-70B SLLM / Safe / PyT (s)	10.3 / 48.0 / 84.0	10.3 / 48.0 / 84.0	0.00	cold=2183.5 W, idle=724.6 W
ServerlessLLM Fig 6a	OPT-13B SLLM / Safe / PyT (s)	2.0 / 8.2 / 14.0	2.0 / 8.2 / 14.0	0.00	cold=545.9 W, idle=181.2 W
Li et al. (2026) Sweep	Conv 1h 0.25x RPS (B2 vs B1 J/tok)	2.195	2.745	-20.01	TTFT +11.4%, TPOT -7.8%
Li et al. (2026) Sweep	Conv 1h 0.50x RPS (B2 vs B1 J/tok)	1.138	1.509	-24.57	TTFT +11.5%, TPOT -14.9%
`Li2026` (1.00x RPS)	Conv 1h 1.00x RPS (B2 vs B1 J/tok)	0.813	1.163	-30.10	TTFT +11.7%, TPOT -19.4%
`Li2026` (1.50x RPS)	Conv 1h 1.50x RPS (B2 vs B1 J/tok)	0.628	0.903	-30.40	TTFT +25.7%, TPOT -26.2%
`Li2026` (2.00x RPS)	Conv 1h 2.00x RPS (B2 vs B1 J/tok)	0.567	0.810	-30.04	TTFT +44.3%, TPOT -31.6%

S3. Complete 24-Hour Trace-Tiling Sensitivity Ablation

Table 3 reports all 12 policy-by-tiling combinations across four synthetic 24-hour trace construction modes (10 paired seeds each = 120 runs total) from `tiling_ablation_summary.csv`. Across all four regimes, `P4` operational SCI remains bounded within 0.0314–0.0320 g/1k tok (maximum relative shift −2.02%) and SLO violation rates remain within 0.817%–0.835%, confirming that bootstrap block reuse does not artificially create the 60.7% carbon advantage over `B1`.

Table 3: Full 24h Trace-Tiling Sensitivity Matrix (10 Paired Seeds per Cell)

Tiling Mode	Policy	SCI_{op} (g/1k)	SCI_{tot} (g/1k)	SLO (%)	P95 TTFT (ms)	Mean Reqs
`orig_280s_repl`	`B1`	0.0815 ± 0.0097	0.0906 ± 0.0097	0.0004 ± 0.0003	268.6	476,119.6
`orig_280s_repl`	`B2`	0.0573 ± 0.0068	0.0663 ± 0.0068	0.1996 ± 0.0157	300.3	476,119.6
`orig_280s_repl`	`P4`	0.0320 ± 0.0061	0.0411 ± 0.0061	0.8237 ± 0.0386	504.5	476,119.6
`perm_280s_norepl`	`B1`	0.0806 ± 0.0094	0.0896 ± 0.0094	0.0008 ± 0.0003	268.7	478,841.8
`perm_280s_norepl`	`B2`	0.0567 ± 0.0066	0.0656 ± 0.0066	0.2164 ± 0.0086	300.6	478,841.8
`perm_280s_norepl`	`P4`	0.0316 ± 0.0059	0.0406 ± 0.0059	0.8174 ± 0.0291	511.2	478,841.8
`fine_90s_perm`	`B1`	0.0807 ± 0.0094	0.0896 ± 0.0094	0.0008 ± 0.0005	268.6	478,556.8
`fine_90s_perm`	`B2`	0.0567 ± 0.0066	0.0657 ± 0.0066	0.2117 ± 0.0094	300.6	478,556.8
`fine_90s_perm`	`P4`	0.0317 ± 0.0059	0.0406 ± 0.0059	0.8202 ± 0.0333	510.9	478,556.8
`coarse_900s_perm`	`B1`	0.0801 ± 0.0091	0.0890 ± 0.0091	0.0008 ± 0.0003	268.7	482,902.5
`coarse_900s_perm`	`B2`	0.0564 ± 0.0065	0.0652 ± 0.0064	0.2240 ± 0.0098	300.9	482,902.5
`coarse_900s_perm`	`P4`	0.0314 ± 0.0057	0.0402 ± 0.0057	0.8351 ± 0.0278	514.9	482,902.5